

Trust Computer E-commerce Web App

**Prepared for: Shiblu Ahmed
Trust Computer-Moulvibazar**

Prepared by: Amdads Group

Proposed platform: Next.js / React • Currency: BDT (15,000)

1. Executive Summary

This document presents the proposed scope and implementation plan for a responsive e-commerce web application for Trust Computer-Moulvibazar. The platform will allow customers to browse computer products, laptops, accessories and CCTV equipment, submit orders, choose supported payment methods, and contact the business through WhatsApp. A secure administration area will support product, inventory, order, payment and reporting workflows.

The plan is based on the information shared for project planning. Product range, operational policies, payment provider approval, delivery terms and final branding assets will be confirmed with the client before implementation.

2. Client & Business Details

Item	Details
Business	Trust Computer-Moulvibazar
Project owner / client contact	Shiblu Ahmed
Address provided	T.S Plaza (2nd Floor), Kusumbagh, Moulvibazar, Bangladesh
Phone provided	01753-765372
Email provided	trustcomputermb@gmail.com
Website listed	https://trustcomputermb.com/
Facebook page	https://www.facebook.com/TrustComputerr/
Business focus stated	Computers and CCTV cameras

3. Project Objectives

- Establish a professional online storefront accessible on mobile, tablet and desktop.
- Present products with images, prices, specifications, availability and warranty details where supplied.
- Enable customers to create and submit online orders.
- Support cash on delivery and an approved online payment gateway.
- Provide a secure admin dashboard for daily business operations.
- Manage stock quantities and maintain an auditable inventory movement history.
- Provide sales, order and inventory reports based on recorded transactions.
- Make customer inquiries easier through WhatsApp and direct contact options.

4. Project Scope & Features

4.1 Customer storefront

- Homepage with promotional banner, categories, featured products and business information.
- Product listing with search, category/brand filters, price sorting and availability indicators.
- Product details with gallery, model/SKU, specifications, price, stock, warranty and related items.
- Shopping cart with quantity changes, item removal, subtotal and delivery estimate.
- Guest checkout with customer name, phone, delivery address, payment choice and order summary.
- Order confirmation and secure order tracking.
- Contact/about pages, policies, store location and WhatsApp click-to-chat.

4.2 Admin dashboard

- Dashboard summary for orders, sales, pending orders and stock alerts.

- Product and category management, including images, descriptions, SKUs, prices and product status.
- Inventory management with stock-in, stock-out, adjustments, low-stock thresholds and movement history.
- Order management with status updates, customer details, payment status, notes and delivery information.
- Payment records, transaction references, failed/pending states and refund tracking where supported.
- Reports with date filters and CSV export for sales, orders and inventory.
- Admin/staff roles and audit records for sensitive changes.

4.3 WhatsApp

A click-to-chat button can open WhatsApp with a prefilled product or order inquiry. Automated transactional messages through the WhatsApp Business Platform are a separate optional integration and require appropriate business setup, customer consent and approved messaging templates.

4.4 Payments

Cash on delivery is proposed as a baseline option. One online payment gateway will be selected after the client confirms merchant eligibility, fees and supported methods. Potential providers to evaluate include SSLCOMMERZ and approved direct integrations for local payment services. Provider availability and approval are not assumed.

- Payment confirmation must be verified server-side using the provider's supported process.
- Never store card numbers or sensitive payment credentials in the application database.
- Track order status separately from payment status.
- Handle pending, failed, cancelled, successful and refunded payment states.

5. Proposed Pages & Routes

Area	Page / Route	Purpose
Store	/	Homepage and featured products
Store	/products	Searchable product catalog
Store	/products/[slug]	Product details
Store	/categories/[slug]	Category product listing
Store	/cart	Shopping cart
Store	/checkout	Checkout and order placement
Store	/order-confirmation/[orderNumber]	Order confirmation
Store	/track-order	Secure order tracking
Store	/about, /contact	Business information and contact
Store	/privacy-policy, /terms, /delivery-policy, /return-policy	Customer-facing policies
Admin	/admin	Dashboard
Admin	/admin/products	Product management
Admin	/admin/categories	Categories and brands
Admin	/admin/inventory	Stock management
Admin	/admin/orders	Order management
Admin	/admin/payments	Payment records
Admin	/admin/reports	Sales and inventory reports
Admin	/admin/settings	Business and delivery settings

6. UI/UX & Branding Direction

The proposed visual direction follows the blue, red and white palette suggested by the supplied Trust Computer logo. Final colors, logo files and typography should be approved by the client.

Design element	Proposed direction
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Primary color	Blue #263E9A
Accent color	Red #E5222B
Background	White #FFFFFF with light-gray sections #F3F5F9
Typography	Readable sans-serif; support Bengali and English content
Layout	Mobile-first, responsive, clear navigation and prominent search
Product cards	Consistent images, readable pricing, stock state and clear action
Accessibility	Keyboard support, meaningful labels, adequate contrast and visible focus states

7. Technology & Architecture

Layer	Proposed technology / role
Application	Next.js App Router + React + TypeScript
Styling/UI	Tailwind CSS and reusable accessible UI components
Database	PostgreSQL
Data access	Prisma ORM and versioned migrations
Validation	Zod; server-side validation for all sensitive actions
Authentication	Secure admin authentication with role-based authorization
Images	Cloud/object storage or managed image service
Charts	Recharts or equivalent reporting visualization
Testing	Unit/integration tests plus Playwright end-to-end tests
Deployment	Vercel or compatible Node.js hosting, subject to deployment needs

Recommended approach: a modular monolithic application for the initial release, with clearly separated product, inventory, order, payment and reporting modules.

8. Database Overview

The final database schema will be confirmed during technical design. Core entities are expected to include:

Entity	Purpose / key information
User	Customer/admin identity, contact and role
Category	Hierarchical product categories
Brand	Product brand records
Product	Catalog item, description, status and category
ProductVariant	SKU, price, attributes, stock threshold and active state
ProductImage	Image URL, alt text and ordering
Order	Order number, customer/address snapshot, totals and statuses
OrderItem	Product/SKU/name/price/quantity snapshots
Payment	Provider, transaction reference, amount and payment state
InventoryMovement	Stock change, reason, reference, actor and timestamp
StockReservation	Temporary stock holds during checkout/payment
OrderStatusHistory	Timestamped order status changes
PaymentWebhookEvent	Idempotency and webhook processing records
AdminAuditLog	Sensitive admin actions and actor
DeliveryMethod / Settings	Configurable delivery fees and business settings
Return / Refund	Return authorization and refund tracking, if included

Data integrity requirements: unique SKUs and order numbers, decimal monetary fields, database transactions for checkout, immutable order-item snapshots, and safe stock updates to prevent overselling.

9. Core Workflows

9.1 Order placement

1. Customer selects products and proceeds to checkout.
2. Server reloads current prices and verifies availability; browser-submitted totals are not trusted.
3. Server calculates subtotal, discounts, delivery fee and grand total.
4. Order and order-item snapshots are created transactionally; stock is reserved or deducted according to the agreed policy.
5. Cash-on-delivery orders enter the appropriate confirmation workflow; online orders create a payment attempt.
6. Payment confirmation is verified server-side before marking the payment as paid.
7. Admin processes, dispatches and completes the order; status history is retained.

9.2 Inventory control

- Use stock movements for receipts, sales, adjustments and approved returns.
- Set low-stock thresholds per variant/SKU.
- Use atomic updates or database locking for concurrent purchases.
- Expire abandoned reservations according to an agreed duration.
- Restore stock only when the order previously reserved/deducted it; inspect returns before restocking.

10. Reports & Business Metrics

- Sales by day, week and month; gross sales, discounts and net sales based on agreed definitions.
- Order counts and breakdown by status.
- Average order value and sales by product/category.
- Payment method and payment outcome summaries.
- Current stock, low-stock/out-of-stock lists and stock movement history.
- CSV export with appropriate admin permissions.

Reports will reflect recorded application data. An order placed is not automatically equivalent to payment collected; reporting definitions will be agreed with the client.

11. Security, Privacy & Reliability

- Server-side authentication and authorization for every admin operation.
- Secure password/session handling, rate limits and protection against common web vulnerabilities.
- Validate and normalize customer inputs; restrict access to personal information.
- Verify payment callbacks/webhooks and prevent duplicate processing.
- Keep secrets in deployment environment variables, never in client-side code.
- Use database backups, error monitoring, structured logs and a tested restore process.
- Publish privacy, terms, delivery and return policies before launch.
- Use secure order tracking that does not expose private order details through guessable IDs.

12. SEO & Performance

- Unique product/category titles and descriptions, canonical URLs and clean slugs.
- XML sitemap, robots.txt, breadcrumbs and relevant product structured data.
- Optimized responsive images, lazy loading and font optimization.
- Public catalog pages can be cached appropriately; private account/order data must not be publicly cached.
- Accurate local business information and map/directions details, subject to client confirmation.
- Measure performance and accessibility before launch.

13. Implementation Roadmap

Phase	Indicative duration	Deliverables
1. Discovery & requirements	3–5 days	Confirm products, delivery/payment rules, policies, scope and acceptance criteria
2. UI/UX & project setup	4–6 days	Approved wireframes/design system, Next.js setup, database and deployment environments
3. Catalog & storefront	5–8 days	Homepage, categories, search, product details and product admin
4. Cart, checkout & orders	5–8 days	Cart, guest checkout, order creation, status workflow and tracking
5. Inventory & payments	5–8 days	Stock ledger/reservations, gateway integration and payment verification
6. Reports & WhatsApp	3–5 days	Reports, exports, click-to-chat and agreed notifications
7. QA, content & launch	5–7 days	Testing, data entry, security review, training and production launch

Indicative total: approximately 6–10 weeks for one experienced full-stack developer. Actual timing depends on scope approval, content readiness, payment onboarding, feedback cycles and integration requirements.

14. Client Inputs Required

- Final logo files and brand preferences.
- Product spreadsheet with names, SKUs, prices, stock, images, specifications and warranty details.
- Confirmed categories and brands.
- Delivery areas, fees, dispatch process and estimated delivery language.
- Return, exchange, warranty and cancellation policies.
- Preferred payment gateway and merchant onboarding details.
- WhatsApp business number and decision on click-to-chat versus automated messaging.
- Admin users and access roles.
- Domain/hosting ownership and any existing website requirements.
- Approval of page designs, content and launch checklist.

15. Deliverables

- Responsive customer storefront.
- Admin dashboard for catalog, inventory, orders, payments and reports.
- Database schema and migrations.

- Payment integration for the agreed provider, subject to merchant access.
- WhatsApp click-to-chat integration.
- Basic technical SEO and performance optimization.
- Testing and deployment configuration.
- Admin handover/training and technical documentation.

16. Assumptions & Exclusions

- This document defines a proposed scope; it is not a final fixed-price contract.
- Product content, images, prices, warranty claims and policy text will be supplied or approved by the client.
- Payment provider approval, transaction fees, SMS/WhatsApp fees, hosting, domain and third-party subscriptions are external dependencies and may incur separate costs.
- Courier API integration, accounting software integration, multi-warehouse support, POS hardware, advanced CRM, loyalty programs and mobile apps are not included unless separately agreed.
- Exact delivery schedule and commercial terms will be confirmed after requirements and dependencies are finalized.

17. Acceptance Criteria

- Customers can browse and search published products on mobile and desktop.
- Checkout validates current prices and stock server-side and creates an order reliably.
- Admin can manage products, stock, orders and supported payment records.
- Stock changes are auditable and concurrent purchases cannot oversell available inventory.
- Payment status is updated only after trusted server-side verification.
- Reports match documented business definitions and selected date ranges.
- Admin routes are protected; customer order information is not exposed publicly.
- Critical user journeys pass agreed functional and end-to-end tests.

18. Approval

By approving this document, the client confirms that the proposed direction and scope may proceed to detailed requirements, UI design and commercial estimation. Any changes to scope, integrations or deliverables will be documented and mutually approved.

For Trust Computer-Moulvibazar	For Amdads Group
Name: Shiblu Ahmed	Name: Amdadur Rahman
Signature: _____	Signature: _____
Date: _____	Date: _____